PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2, IC3, IC5	- 74121 monostable
IC4	- 7408 quad AND gate
D1-D6	- 1N4007 rectifier diode
T1	- BC547 npn transistor
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R3, R5	- 10-kilo-ohm
R2, R4, R6-R8	- 1-kilo-ohm
Capacitors:	
C1, C5, C6, C8, C10	- 0.1µF ceramic disk
C2	- 1000µF, 25V electrolytic
C3	- 0.22µF ceramic disk
C4, C7	- 2.2µF, 16V electrolytic
C9	- 470µF, 16V electrolytic
Miscellaneous:	
X1	- 230V 50Hz AC primary to 9V-0-9V 500mA secondary transformer
RL1	- 5V single change-over relay
CON1-CON3	- 2-pin terminal connector
CON4	- 5-pin connector
S1	- On/off switch
S2	- Push-to-on switch
	- 230V AC bell
	- Digital timer programmable time switch TM-619
	- 230V AC, 50Hz mains supply
J1, J2	- Jumper wire

time, its status changes from 'on' to 'off' and vice versa.

4. Accordingly, the timer clock generates a falling-edge or rising-edge trigger signal for monostable IC2 and IC3. $\overline{Q}$ output of the respective IC generates a low-going pulse with pulse width of 1.5 milliseconds, which, in turn, triggers IC5. Transistor T1 conducts to energise relay RL1 and the bell rings for 3.2 seconds.

5. The process repeats as per the timetable.

Construction and testing

An actual-size PCB layout of the programmable automatic bell system is shown in Fig. 3 and its components layout in Fig. 4.

After mounting components on the PCB, first check $\overline{Q}$ output of IC2 using an oscilloscope; when a positive trigger is given to pin 5 (input B), the pulse width at pin 1 should be 1.5 milliseconds. Similarly, check pulse output pin 1 ($\overline{Q}$) of IC3 by giving a negative trigger to pin 4 (A2 input) of IC3.

The output of IC5 drives the base of npn transistor T1 (BC 547) through diode D6 and resistor R6.

In order to test the bell manually, a push-to-on switch (S2) is provided. It is connected to the base of the transistor through diode D5. When you press S2, transistor T1 conducts to drive relay RL1 and the bell rings continuously as long as S2 is pressed.

In normal condition, transistor T1 gets the pulse from pin 6 of IC5. As it has pulse width of 3.2 seconds, the bell rings only for 3.2 seconds.

After assembling the circuit on the PCB, connect 230V AC to CON1. Also connect the digital timer and 230V AC bell externally to the PCB using external wires. Connect switches S1 and S2, and LED1, on the front panel of the cabinet. Note that the PCB shows only the secondary winding of X1 (9V-0-9V). Affix the switch S1 on the cabinet at a suitable location and connect to primary winding of X1 using external wires. EFY


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